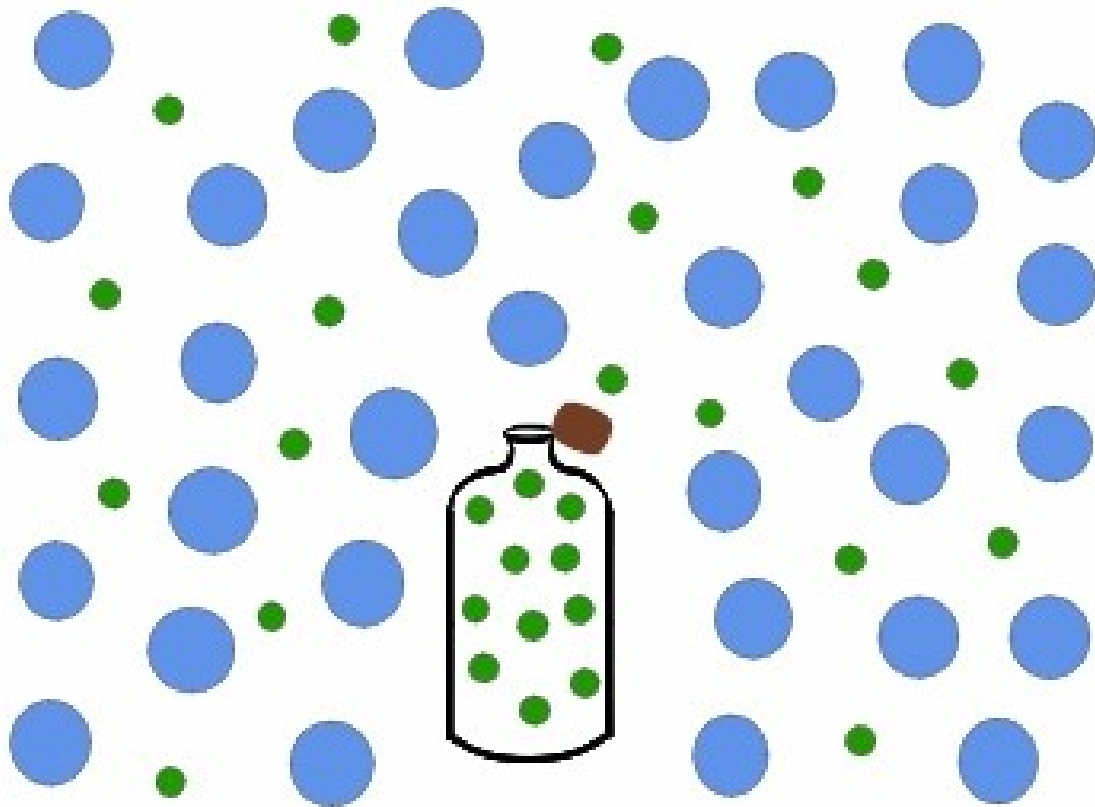


Introduction to DDPM

Denoising Diffusion Probabilistic Models

Group 3

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What Is Diffusion?

- In non-equilibrium statistical physics, the diffusion process refers to:
- --***“The movement of particles or molecules from an area of high concentration to an area of low concentration, driven by a gradient in concentration.”***

What Are Diffusion Models?

In simple terms - ***“Diffusion Models are a class of probabilistic generative models that turn noise to a representative data sample.”***

1. *Unconditional image generation* - converts noise into any “random representative data sample.”
2. *Conditional image generation* - the model is provided additional information via text

A bit of history

Diffusion models in deep learning were first introduced by Sohl-Dickstein et al. in the seminal 2015 paper “Deep Unsupervised Learning using Nonequilibrium Thermodynamics.” Unfortunately, it remained behind the curtains for a while.

But, in 2019, Song et al. published a paper, “*Generative Modeling by Estimating Gradients of the Data Distribution*,” using the same principle but a different approach. In 2020, Ho et al. published the paper, now-popular “*Denoising Diffusion Probabilistic Models*” (DDPM for short).

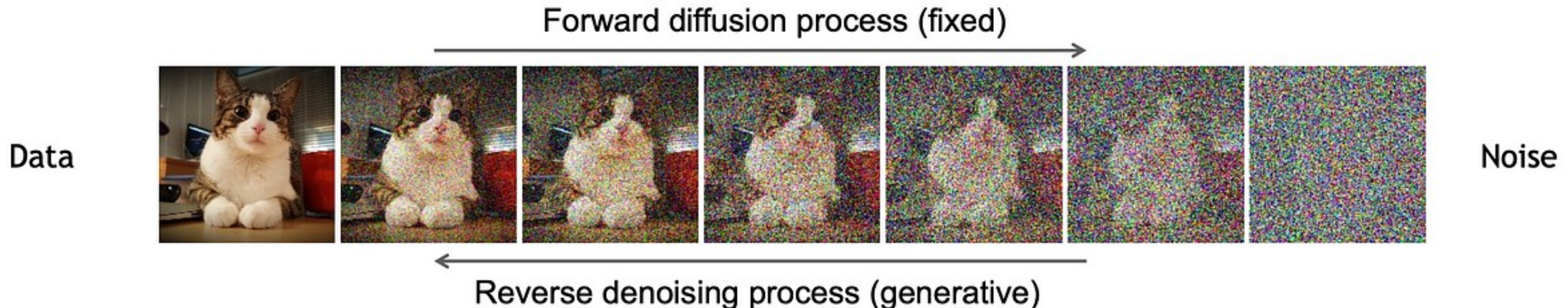
After 2020, research in diffusion models took off . Much progress has been made in creating, training, and improving diffusion-based generative modeling in a relatively short time.

How It Works?

The diffusion method can be summarized as follows:

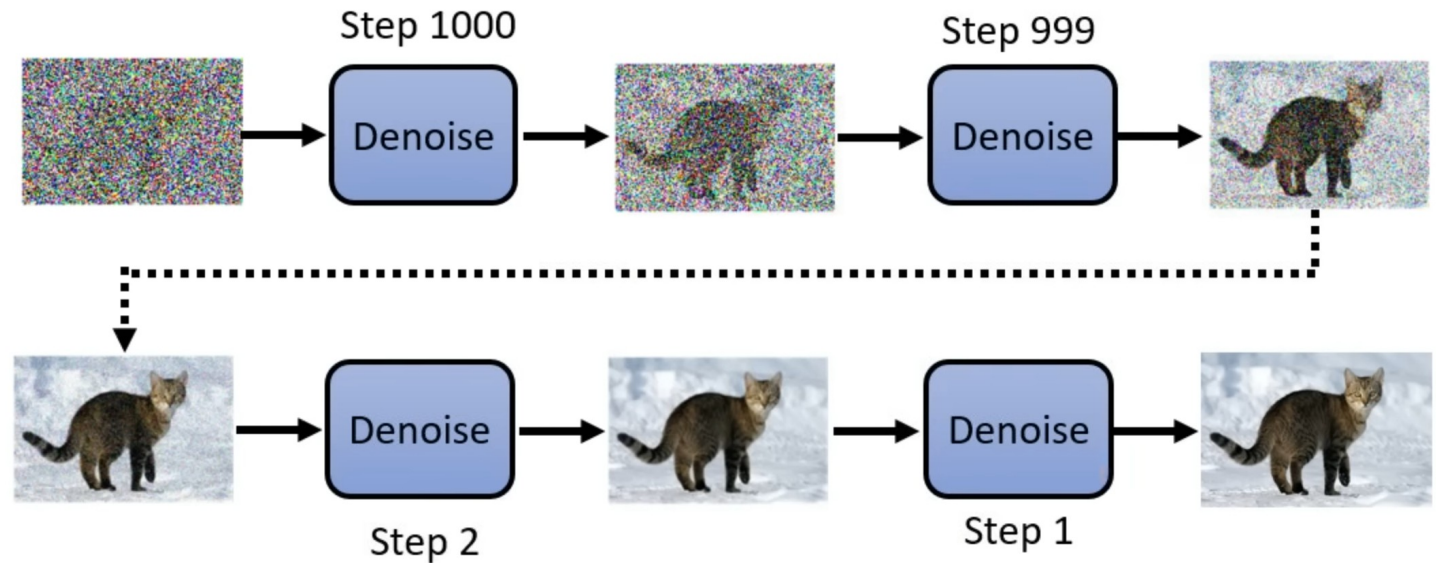
...systematically and slowly destroy the structure in a data distribution through an iterative forward diffusion process. We then learn a reverse diffusion process that restores structure in data, yielding a highly flexible and tractable generative model of the data. This approach allows us to rapidly learn, sample from, and evaluate probabilities in deep generative models ...

- Deep Unsupervised Learning using Nonequilibrium Thermodynamics, 2015



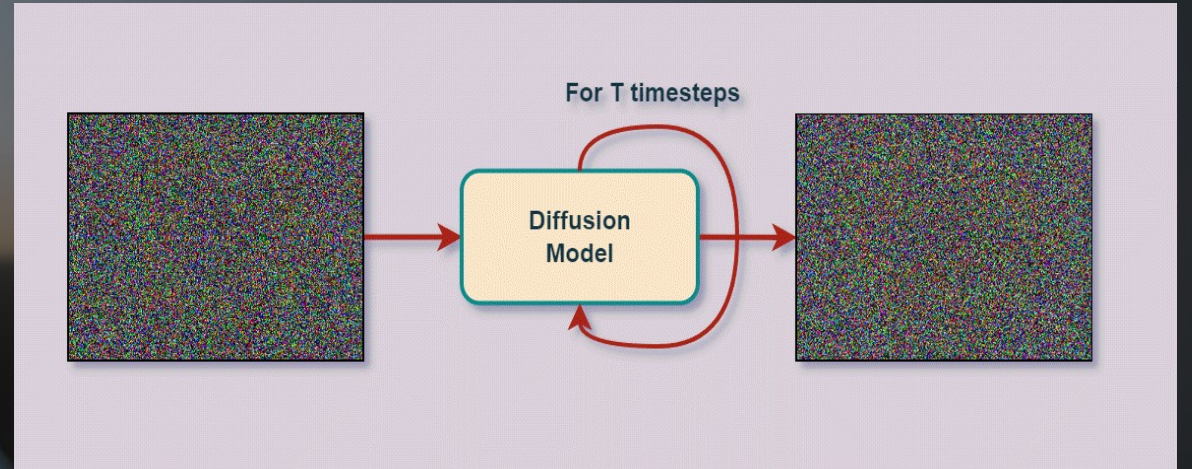
Reverse Process

- Known as denoising
- Smaller step => less noise
- Larger step => more noise
- $p_{\theta}(x_{0:T}) := p(x_T) \prod_{t=1}^T p_{\theta}(x_{t-1}|x_t)$



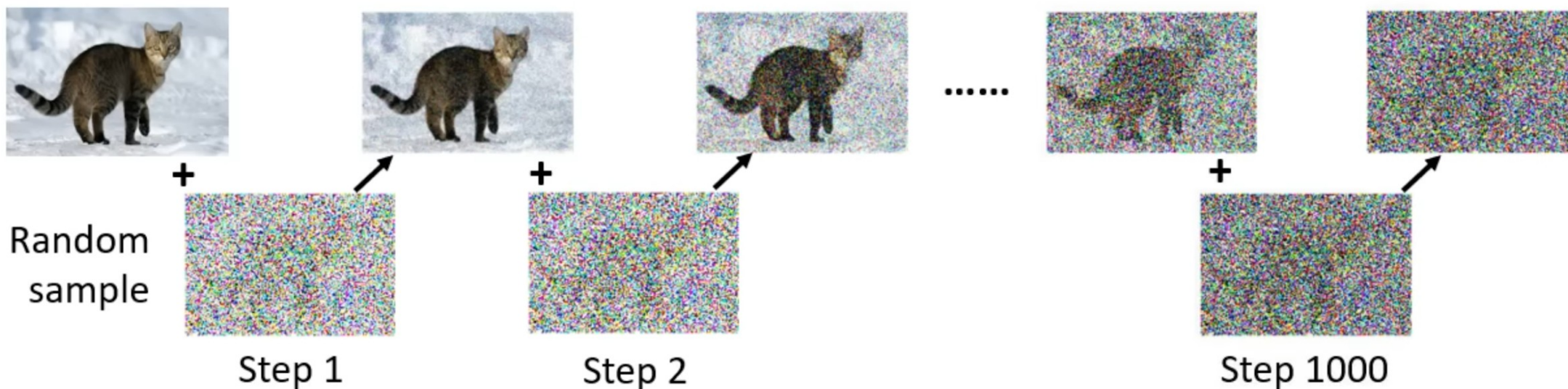
“The sculpture is already complete within the marble block, before I start my work. It is already there, I just have to chisel away the superfluous material.”

— Michelangelo



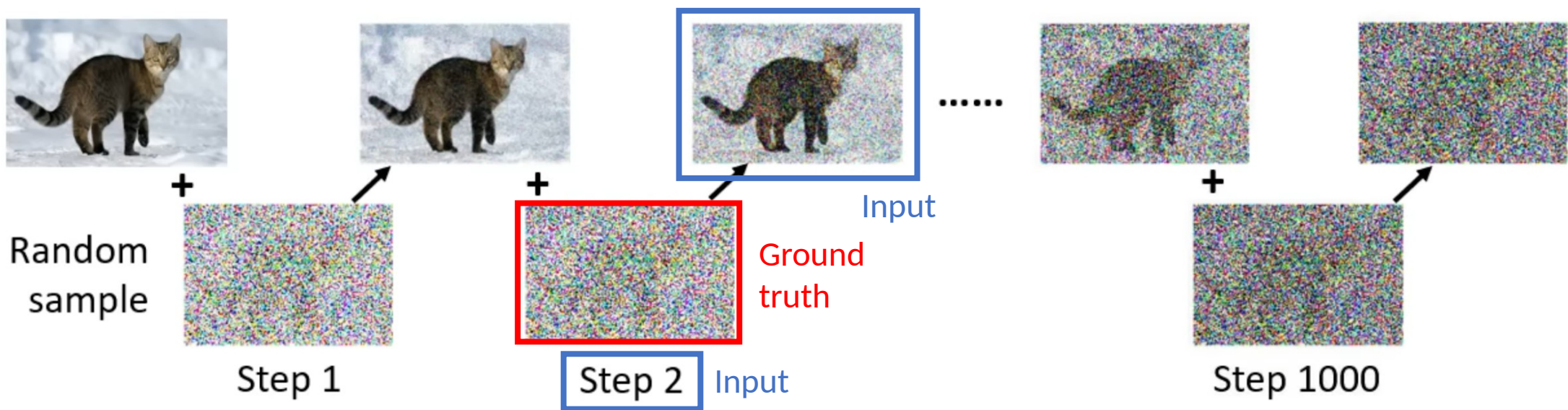
Forward Process

- The original image is slowly corrupted iteratively (a Markov chain) by adding (scaled Gaussian) noise.
- No model is involved at this stage.
- $q(x_{1:T}|x_0) = \prod_{t=1}^T q(x_t|x_{t-1})$



Forward Process

- The original image is slowly corrupted iteratively (a Markov chain) by adding (scaled Gaussian) noise.
- No model is involved at this stage.
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Dall-E 2 by OpenAI

- **Description:** DALL-E 2 is an advanced AI model developed by OpenAI, capable of generating high-quality, creative images from textual descriptions.
- **Innovations:** Create detailed, imaginative, and sometimes whimsical images that can combine unrelated concepts in a coherent visual format.
- **Applications:** Creative industries for conceptual art, advertising, and generating visual ideas from textual descriptions.



Imagen by Google

- **Description:** Imagen, developed by Google, is a state-of-the-art text-to-image AI model. It's renowned for generating photorealistic images from descriptive text, pushing the boundaries of AI's capability in understanding and rendering detailed visuals.
- **Innovations:** Fidelity and resolution in image generation, as well as its ability to handle nuanced and complex descriptions.
- **Applications:** Marketing, content creation, and education.



Stable Diffusion by StabilityAI

- **Description:** Stable Diffusion is an AI model by StabilityAI designed for generating high-quality images from text prompts.
- **Innovations:** Balance between image quality and computational efficiency.
- **Applications:** Aids in democratizing AI art creation, allowing artists, educators.



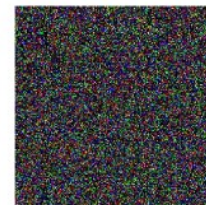
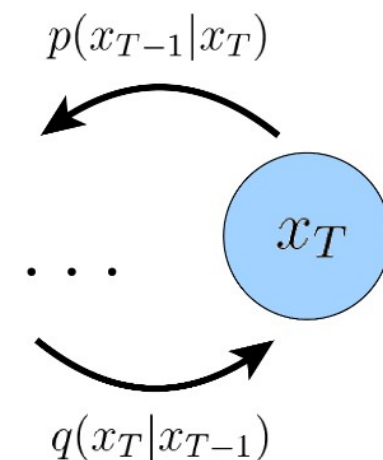
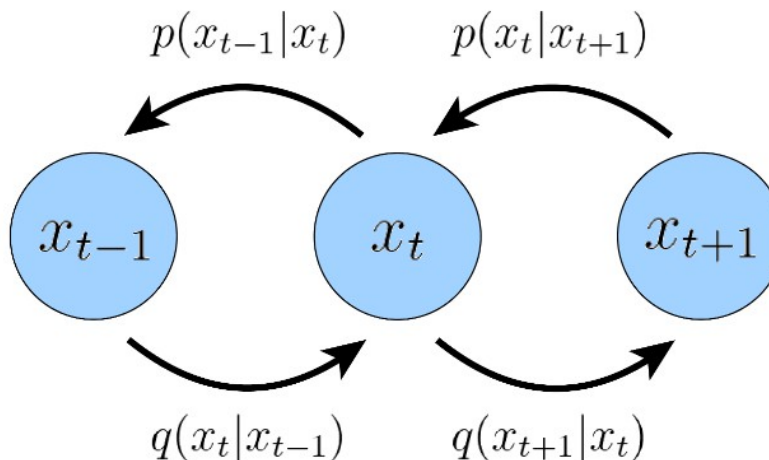
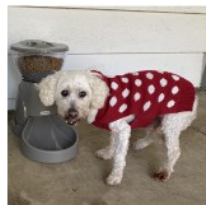
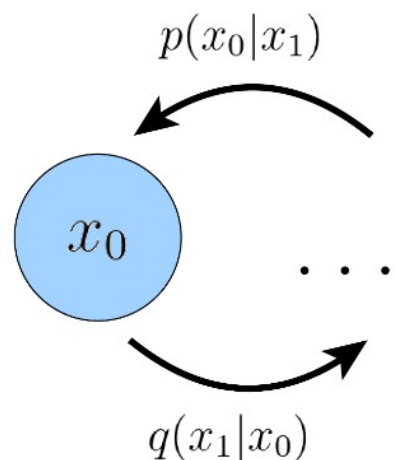
Midjourney

- **Description:** Midjourney is an AI tool that specializes in creating artistic and visually striking images from text descriptions.
- **Innovations:** Artistic styles and abstract imagery, offering a different perspective on AI-generated art.
- **Applications:** Artists, designers, and creatives seeking innovative and unconventional visual interpretations of their ideas.



Notation

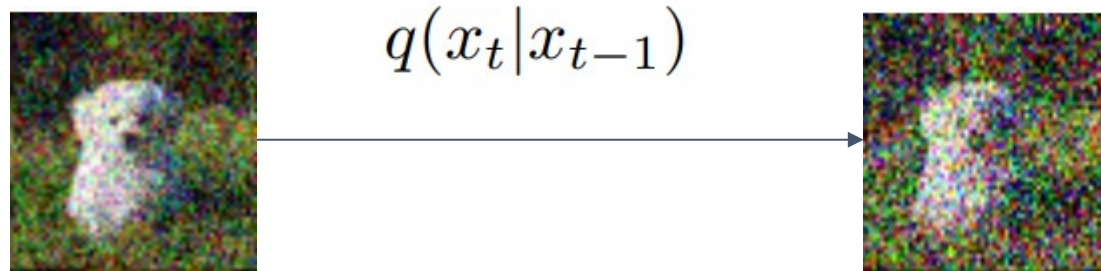
- x_t : object on timestep t
- q : forward diffusion process
- p : reverse diffusion process



Forward process

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I)$$

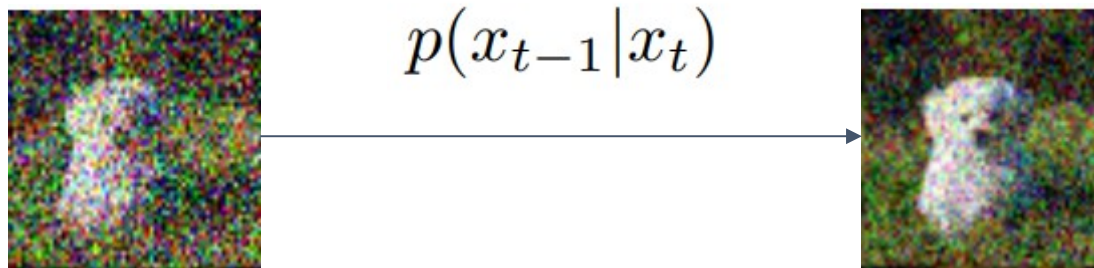
- Distribution of a less noised object given by a more noised object
- Define variance beta ranging from 0 to 1 in a schedule of time step
- Scaling factor to mean is for avoid the variance explode



Reverse process

$$p_{\theta}(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \Sigma_{\theta}(x_t, t))$$

- Distribution of a more noised object given by a less noised object
- Output approximate by a neural network with parameter theta
- Optimize the diffusion model by doing gradient descent on the loss function to optimize the parameter theta



Maximum likelihood estimation

$$\max_{\theta} \log \Pr(x_0)$$

- The probability of x_0 is hard to calculate
- x_0 depends on $x_1, x_2, \dots, x_T$

$$\log p(x_0) = \log \int p(x_0, x_{1:T}) dx_{1:T}$$

Evidence Lower Bound(ELBO)

- ELBO is always smaller than the evidence
- Maximize the ELBO is a proxy objective to maximize the evidence

$$\begin{aligned}\log p(x_0) &= \log \int p(x_0, x_{1:T}) dx_{1:T} \\ &= \log \int \frac{p(x_0, x_{1:T}) q(x_{1:T} | x_0)}{q(x_{1:T} | x_0)} dx_{1:T} \\ &= \log \mathbb{E}_{q(x_{1:T} | x_0)} \left[\frac{p(x_0, x_{1:T})}{q(x_{1:T} | x_0)} \right] \\ &\leq \mathbb{E}_{q(x_{1:T} | x_0)} \left[\log \frac{p(x_0, x_{1:T})}{q(x_{1:T} | x_0)} \right] = \mathbb{E}_{q(x_{1:T} | x_0)} \left[\log \frac{p(x_0) \prod_{t=1}^T p_\theta(x_{t-1} | x_t)}{\prod_{t=1}^T q(x_t | x_{t-1})} \right]\end{aligned}$$

Evidence Lower Bound(ELBO)

$$ELBO = \mathbb{E}_{q(x_{1:T}|x_0)} \left[\log \frac{p(x_0, x_{1:T})}{q(x_{1:T}|x_0)} \right]$$

$$= L_1 - \sum_{t=2}^T L_t - L_{T+1}$$

$$L_1 := \log p_{\theta}(x_0|x_1)$$

$$L_t := D_{KL}(q(x_{t-1}|x_t, x_0) || p_{\theta}(x_{t-1}|x_t))$$

$$L_{T+1} := D_{KL}(q(x_T|x_0) || p(x_T))$$

- L_1 : Reconstruction term
- L_t : Denoising matching term
- L_{T+1} : Prior matching term

Reconstruction term

$$L_1 := \log p_{\theta}(x_0|x_1)$$

- Predict the log probability of the original data sample given the first-step noised object.
- It can be approximated and optimized using a Monte Carlo estimate.

Prior matching term

$$L_{T+1} := D_{KL}(q(x_T|x_0)||p(x_T))$$

- The KL divergence between two distribution
- Represents how close the distribution of the final noisified input is to the standard Gaussian prior
- It is equal to zero(two distributions are the same) if we learn a perfect model

Rewrite forward process

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I)$$

Define:

$$\alpha_t = 1 - \beta_t$$

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$$

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_{t-1}, (1 - \alpha_t)I)$$

Sample object from forward process

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I)$$

reparameterization trick:

$$x \sim \mathcal{N}(x; \mu, \sigma^2)$$

$$x = \mu + \sigma\epsilon, \quad \epsilon \sim \mathcal{N}(0, 1)$$

$$\begin{aligned} x_t &= \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon_{t-1} \\ &= \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{\alpha_t(1 - \alpha_{t-1})}\epsilon_{t-2} + \sqrt{1 - \alpha_t}\epsilon_{t-1} \\ &= \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{\sqrt{\alpha_t - \alpha_t\alpha_{t-1}}^2 + \sqrt{1 - \alpha_t}^2}\epsilon_{t-2} \\ &= \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{1 - \alpha_t\alpha_{t-1}}\epsilon_{t-2} \\ &\dots\dots \\ &= \sqrt{\alpha_t\alpha_{t-1}\dots\alpha_0}x_0 + \sqrt{1 - \alpha_t\alpha_{t-1}\dots\alpha_0}\epsilon_0 \\ &= \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon_0 \end{aligned}$$

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$$

Denoising matching term

$$L_t := D_{KL}(q(x_{t-1}|x_t, x_0) || p_{\theta}(x_{t-1}|x_t))$$

- Use the neural network to learn the reverse diffusion process p_{θ} to approximate the ground-truth denoising process q
- This term is minimized when the two distributions match as closely as possible, as measured by their KL Divergence.

Ground-truth denoising process

$$\begin{aligned} q(x_{t-1}|x_t, x_0) &= \frac{q(x_t|x_{t-1}, x_0)q(x_{t-1}|x_0)}{q(x_t|x_0)} \\ &= \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)} \\ &= \frac{\mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I)\mathcal{N}(x_t; \sqrt{\bar{\alpha}_{t-1}}x_0, (1 - \bar{\alpha}_{t-1})I)}{\mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)} \\ &\propto \mathcal{N}(x_{t-1}; \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})x_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)x_0}{1 - \bar{\alpha}_t}, \frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}I) \end{aligned}$$

- Mean is a formula of x_t and x_0
- Variance is irrelevant to object x

Loss function

$$\begin{aligned} & \arg \min_{\theta} D_{KL}(q(x_{t-1}|x^t, x_0) || p_{\theta}(x_{t-1}|x_t)) \\ &= \arg \min_{\theta} D_{KL}(\mathcal{N}(x_{t-1}; \mu_q, \Sigma) || \mathcal{N}(x_{t-1}; \mu_{\theta}, \Sigma)) \\ &= \arg \min_{\theta} \frac{1}{2\sigma^2} \left[||\mu_{\theta} - \mu_q||_2^2 \right] \end{aligned}$$

mean squared error between the actual mean and predicted mean of denoising process

Reformulation

Recall the formula of sampling x_t from x_0 :

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon_0$$
$$x_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_0}{\sqrt{\bar{\alpha}_t}}$$

Reformulate the mean of ground-truth denoising process and the reverse diffusion

q

$$\mu_q = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})x_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)x_0}{1 - \bar{\alpha}_t}$$
$$= \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}}\epsilon_0$$

$$\mu_\theta = \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}}\epsilon_\theta(x_t, t)$$

Loss function

$$\begin{aligned} & \arg \min_{\theta} D_{KL}(q(x_{t-1}|x^t, x_0)||p_{\theta}(x_{t-1}|x_t)) \\ &= \arg \min_{\theta} D_{KL}(\mathcal{N}(x_{t-1}; \mu_q, \Sigma)||\mathcal{N}(x_{t-1}; \mu_{\theta}, \Sigma)) \\ &= \arg \min_{\theta} \frac{1}{2\sigma^2} \left[||\mu_{\theta} - \mu_q||_2^2 \right] \\ &= \arg \min_{\theta} \frac{(1 - \alpha_t)^2}{2\sigma^2 \alpha_t (1 - \bar{\alpha}_t)} \left[||\epsilon_{\theta}(x_t, t) - \epsilon_0||_2^2 \right] \end{aligned}$$

mean squared error between the actual noise and predicted noise

Summary

- Diffusion process: adding noise and denoising
- Maximum likelihood estimation
- Evidence lower bound
- Loss function